

Operating System : Scheduling Algorithm

Different Time Concepts in CPU Scheduling

Arrival Time:

In CPU Scheduling, the arrival time refers to the moment in time when a process enters the ready queue and is awaiting execution by the CPU.

Burst Time:

Burst time, also referred to as “execution time”. It is the amount of CPU time the process requires to complete its execution.

It's a duration of time.

Completion Time:

Completion time is when a process finishes execution and is no longer being processed by the CPU. It is the summation of the arrival, waiting, and burst times.

Different Time Concepts in CPU Scheduling

Turnaround Time:

The time elapsed between the arrival of a process and its completion is known as turnaround time. That is, the duration it takes for a process to complete its execution and leave the system.

$$\text{Turnaround Time} = \text{Completion Time} - \text{Arrival Time}$$

Waiting Time:

This is a process's duration in the ready queue before it begins executing.

$$\text{Waiting Time} = \text{Turnaround Time} - \text{Burst Time}$$

Response Time:

Response time is the amount of time it takes for the CPU to respond to a request made by a process. It is the duration between the arrival of a process and the first time it runs.

$$\text{Response Time} = \text{Time it Started Executing} - \text{Arrival Time}$$

Scheduling Algorithm

- Basic Concepts
- First-Come, First-Served (FCFS) Scheduling Algorithm
- Shortest-Job-First (SJF) Scheduling Algorithm
 - Nonpreemptive SJF Scheduling
 - Preemptive SJF Scheduling
- Priority Scheduling Algorithm
 - Nonpreemptive Scheduling Algorithm
 - Preemptive Scheduling Algorithm
- Round-Robin Scheduling Algorithm
- Advantages of Round-Robin Scheduling Algorithm
- Disadvantages of Round-Robin Scheduling Algorithm

Scheduling Algorithm

- The scheduling algorithm deals with the problem of deciding which of the process in the ready queue is to be allocated the CPU.
- There are many CPU scheduling algorithm as shown below.
 - First-Come, First- Served (FCFS) scheduling algorithm
 - Shortest -Job –First (SJF) scheduling algorithm
 - Priority scheduling algorithm
 - Round- Robin scheduling algorithm

FCFS Scheduling Algorithm

- First-Come, First-Served Scheduling Algorithm is nonpreemptive algorithm.
- It is the simplest of all the scheduling algorithms.
- The key concept of this algorithm is “The process which comes first in the ready queue will allocate the CPU first”.
- The next process will allocate the CPU only after the previous gets fully executed.

FCFS Scheduling Algorithm

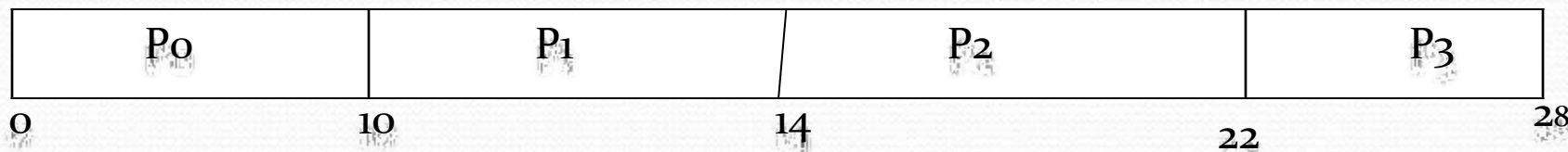
Example- 1:

Sr. No	Process	Execution Time/ Burst Time(ms)
1	p0	10
2	p1	4
3	p2	8
4	P ₃	6

If the arrival time is not given. We can assume arrival time of all process is 0.

The CPU executes them in the order they are listed (or the order specified in the problem).

Gantt Chart:



FCFS Scheduling Algorithm

Completion Time:

$P0 = 10, P1 = 14, P2 = 22, P3 = 28$

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT)

Turnaround Time(TAT) of each process

$P0 : 10 - 0 = 10 \text{ ms}$

$P1 : 14 - 0 = 14 \text{ ms}$

$P2 : 22 - 0 = 22 \text{ ms}$

$P3 : 28 - 0 = 28 \text{ ms}$

Average Turnaround Time(ATAT) = Total turn around time of all processes / Total no of processes

Average Turnaround Time(ATAT) = $(10+14+22+28) / 4 = 18.5 \text{ ms}$

FCFS Scheduling Algorithm

Waiting Time(WT) = Turnaround Time(TAT) – Burst Time(BT)

Waiting Time(WT) of each process

$$P0 : 10 - 10 = 0 \text{ ms}$$

$$P1 : 14 - 4 = 10 \text{ ms}$$

$$P2 : 22 - 8 = 14 \text{ ms}$$

$$P3 : 28 - 6 = 22 \text{ ms}$$

Average Waiting Time(AWT) = Total waiting time of all processes / Total no of processes

$$\text{Average Waiting Time(AWT)} = (0+10+14+22) / 4 = 11.5 \text{ ms}$$

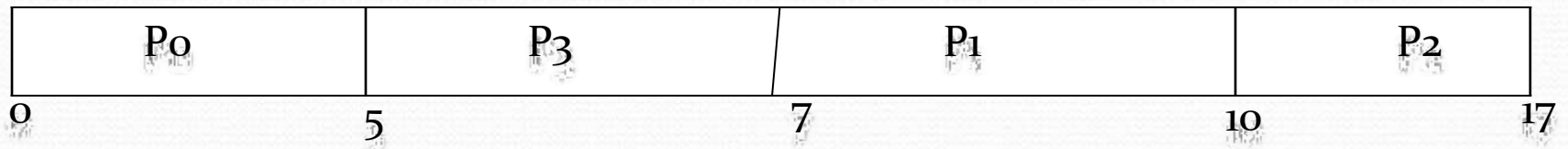
FCFS Scheduling Algorithm

Example- 2:

Sr. No	Process	Arrival Time	Execution Time/ Burst Time(ms)
1	p0	0	5
2	p1	4	3
3	p2	6	7
4	p3	2	2

When Arrival Time is given

Gantt Chart:



FCFS Scheduling Algorithm

Completion Time:

$P0 = 5, P1 = 10, P2 = 17, P3 = 7$

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT)

Turnaround Time(TAT) of each process

$P0 : 5 - 0 = 5 \text{ ms}$

$P1 : 10 - 4 = 6 \text{ ms}$

$P2 : 17 - 6 = 11 \text{ ms}$

$P3 : 7 - 2 = 5 \text{ ms}$

Average Turnaround Time(ATAT) = Total turn around time of all processes / Total no of processes

Average Turnaround Time(ATAT) = $(5+6+11+5) / 4 = 6.75 \text{ ms}$

FCFS Scheduling Algorithm

$$\text{Waiting Time(WT)} = \text{Turnaround Time(TAT)} - \text{Burst Time(BT)}$$

Waiting Time(WT) of each process

$$P0 : 5 - 5 = 0 \text{ ms}$$

$$P1 : 6 - 3 = 3 \text{ ms}$$

$$P2 : 11 - 7 = 4 \text{ ms}$$

$$P3 : 5 - 2 = 3 \text{ ms}$$

$$\text{Average Waiting Time(AWT)} = \text{Total waiting time of all processes} / \text{Total no of processes}$$

$$\text{Average Waiting Time(AWT)} = (0+3+4+3) / 4 = 2.5 \text{ ms}$$

FCFS Scheduling Algorithm (Convoy Effect)

<u>Process</u>	<u>Burst Time</u>
P_1	24
P_2	3
P_3	3

- Suppose that the processes arrive in the order: P_1, P_2, P_3
The Gantt Chart for the schedule is:



- Waiting time for $P_1 = 0$; $P_2 = 24$; $P_3 = 27$
- Average waiting time: $(0 + 24 + 27)/3 = 17$

FCFS Scheduling Algorithm

Suppose that the processes arrive in the order:

$$P_2, P_3, P_1$$

- The Gantt chart for the schedule is:



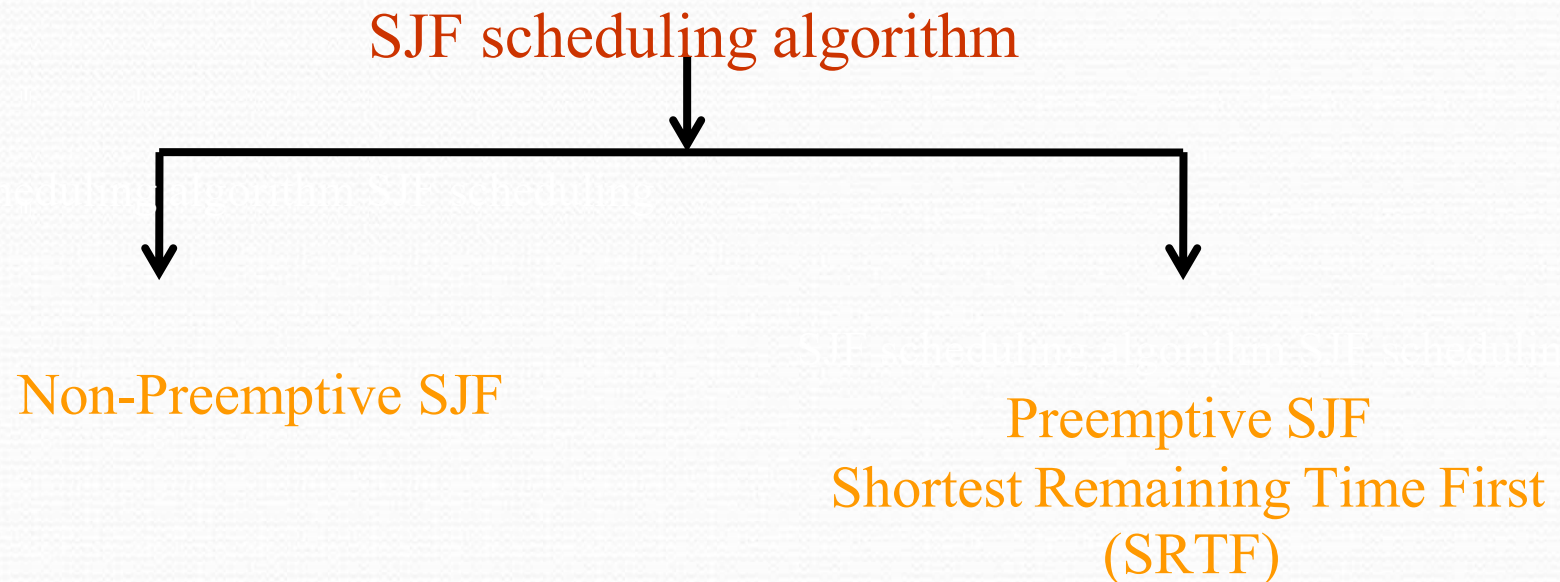
- Waiting time for $P_1 = 6$; $P_2 = 0$; $P_3 = 3$
- Average waiting time: $(6 + 0 + 3)/3 = 3$
- Much better than previous case
- **Convoy effect** - short process behind long process
 - Consider one CPU-bound and many I/O-bound processes

SJF Scheduling Algorithm

- Shortest Job First scheduling(SJF) works on the process with the shortest **burst time**.
- SJF is the best approach to minimize waiting time.
- SJF is a Greedy Algorithm.
- In case of a tie, it is broken by FCFS scheduling algorithm.

Types of SJF Scheduling Algorithm

- SJF scheduling algorithm can be categorized into two parts.



SJF Scheduling Algorithm

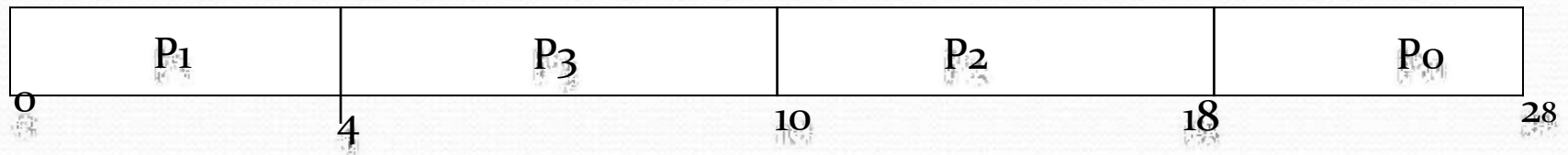
- The SJF algorithm may be either preemptive or non-preemptive
- A preemptive SJF will preempt this currently executing process and starts the execution of newly entered process.
- Nonpreemptive SJF will allow the currently executing process to complete its burst time without any interruption in its execution.
- Preemptive SJF scheduling is sometimes called Shortest-Remaining-Time-First (SRTF) scheduling

Nonpreemptive SJF Scheduling Algorithm

Sr. No	Process	Execution Time/ Burst Time(ms)
1	p0	10
2	p1	4
3	p2	8
4	p3	6

If the arrival time is not given. We can assume arrival time of all process is 0.

Gantt Chart:



Nonpreemptive SJF Scheduling Algorithm

Completion Time:

$P_0 = 28, P_1 = 4, P_2 = 18, P_3 = 10$

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT)

Turnaround Time(TAT) of each process

$P_0 : 28 - 0 = 28 \text{ ms}$

$P_1 : 4 - 0 = 4 \text{ ms}$

$P_2 : 18 - 0 = 18 \text{ ms}$

$P_3 : 10 - 0 = 10 \text{ ms}$

Average Turnaround Time(ATAT) = Total turn around time of all processes / Total no of processes

Average Turnaround Time(ATAT) = $(28+4+18+10) / 4 = 15 \text{ ms}$

Nonpreemptive SJF Scheduling Algorithm

Waiting Time(WT) = Turnaround Time(TAT) – Burst Time(BT)

Waiting Time(WT) of each process

P0 : $28 - 10 = 18$ ms

P1 : $4 - 4 = 0$ ms

P2 : $18 - 8 = 10$ ms

P3 : $10 - 6 = 4$ ms

Average Waiting Time(AWT) = Total waiting time of all processes / Total no of processes

Average Waiting Time(AWT) = $(18 + 0 + 10 + 4) / 4 = 8$ ms

Nonpreemptive SJF Scheduling Algorithm

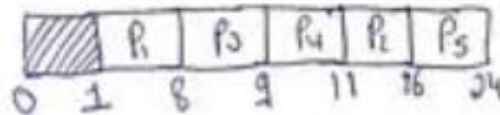
Example 2

Criteria — Burst Time
mode — Non-preemptive

Process	AT	BT	CT	TAT	WT
1	1	7	8	7	0
2	2	5	16	14	2
3	3	1	9	6	5
4	4	2	11	7	5
5	5	8	24	19	11
			53	30	

$$ATAT = \frac{53}{5} \Rightarrow 10.6$$

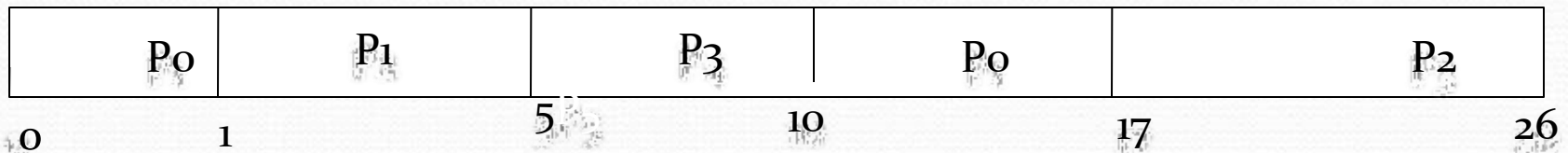
$$AWT = \frac{30}{5} \Rightarrow 6$$



Preemptive SJF Algorithm

Sr. No	Process	Arrival Time	Execution Time/ Burst Time(ms)
1	p0	0	8
2	p1	1	4
3	p2	2	9
4	p3	3	5

Gantt Chart



Preemptive SJF Scheduling Algorithm

Completion Time:

$P_0 = 17, P_1 = 5, P_2 = 26, P_3 = 10$

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT)

Turnaround Time(TAT) of each process

$P_0 : 17 - 0 = 17 \text{ ms}$

$P_1 : 5 - 1 = 4 \text{ ms}$

$P_2 : 26 - 2 = 24 \text{ ms}$

$P_3 : 10 - 3 = 7 \text{ ms}$

Average Turnaround Time(ATAT) = Total turn around time of all processes / Total no of processes

Average Turnaround Time(ATAT) = $(17+4+24+7) / 4 = 13 \text{ ms}$

Preemptive SJF Scheduling Algorithm

Waiting Time(WT) = Turnaround Time(TAT) – Burst Time(BT)

Waiting Time(WT) of each process

P0 : $17 - 8 = 9$ ms

P1 : $4 - 4 = 0$ ms

P2 : $24 - 9 = 15$ ms

P3 : $7 - 5 = 2$ ms

Average Waiting Time(AWT) = Total waiting time of all processes / Total no of processes

Average Waiting Time(AWT) = $(9 + 0 + 15 + 2) / 4 = 6.5$ ms

Time	Process Running	Remaining Times	Notes
0	P0	P0=8	Only P0 has arrived
1	P1	P0=7, P1=4	P1 arrived, has shorter burst
2	P1	P0=7, P1=3, P2=9	P1 still shortest
3	P1	P0=7, P1=2, P2=9, P3=5	P1 still shortest
4	P1	P0=7, P1=1, P2=9, P3=5	P1 still shortest
5	P3	P0=7, P2=9, P3=5	P3 < P0 & P2 now
6	P3	P0=7, P2=9, P3=4	—
7	P3	P0=7, P2=9, P3=3	—
8	P3	P0=7, P2=9, P3=2	—
9	P3	P0=7, P2=9, P3=1	—
10	P0	P0=7, P2=9	P0 < P2
11	P0	P0=6, P2=9	—
12	P0	P0=5, P2=9	—
13	P0	P0=4, P2=9	—
14	P0	P0=3, P2=9	—
15	P0	P0=2, P2=9	—
16	P0	P0=1, P2=9	—
17	P2	P2=9	Only P2 remains
18	P2	P2=8	—
19	P2	P2=7	—
20	P2	P2=6	—
21	P2	P2=5	—
22	P2	P2=4	—
23	P2	P2=3	—
24	P2	P2=2	—
25	P2	P2=1	—
26	P2	P2=0	P2 completes

Priority Scheduling

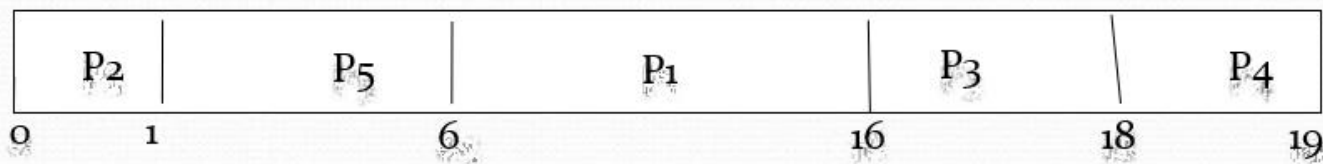
- A priority is associated with each process and the CPU is allocated to the process with the highest priority.
- Equal priority processes are scheduled in FCFS order.
- Priorities are generally some fixed range of numbers such as 0 to 7. some system represents low numbers to represent low priority while others use low numbers to represent high priority.

Priority Scheduling Algorithm

Nonpreemptive Scheduling Algorithm

Sr. No	Process	Burst time	Priority
1	P ₁	10	3
2	P ₂	1	1 (H)
3	P ₃	2	4
4	P ₄	1	5 (L)
5	P ₅	5	2

Gantt Chart:



Priority Scheduling Algorithm

Completion Time:

$P1 = 16, P2 = 1, P3 = 18, P4 = 19, P5 = 6$

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT)

Turnaround Time(TAT) of each process

$P1 : 16 - 0 = 16 \text{ ms}$

$P2 : 1 - 0 = 1 \text{ ms}$

$P3 : 18 - 0 = 18 \text{ ms}$

$P4 : 19 - 0 = 19 \text{ ms}$

$P5 : 6 - 0 = 6 \text{ ms}$

Average Turnaround Time(ATAT) = Total turn around time of all processes / Total no of processes

Average Turnaround Time(ATAT) = $(16+1+18+19+6) / 5 = 12 \text{ ms}$

Priority Scheduling Algorithm

Waiting Time(WT) = Turnaround Time(TAT) – Burst Time(BT)

Waiting Time(WT) of each process

$$P1 : 16 - 10 = 6 \text{ ms}$$

$$P2 : 1 - 1 = 0 \text{ ms}$$

$$P3 : 18 - 2 = 16 \text{ ms}$$

$$P4 : 19 - 1 = 18 \text{ ms}$$

$$P5 : 6 - 5 = 1 \text{ ms}$$

Average Waiting Time(AWT) = Total waiting time of all processes / Total no of processes

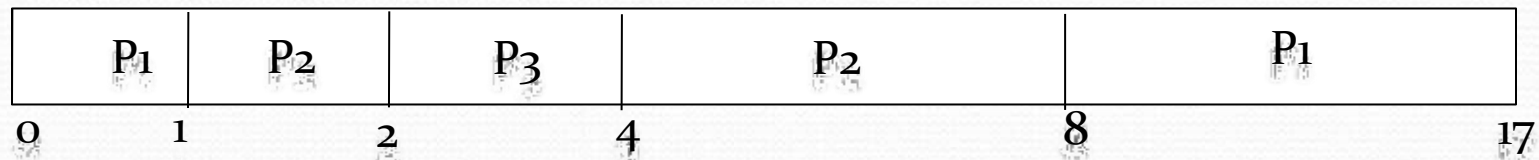
$$\text{Average Waiting Time(AWT)} = (6+0+16+18+1) / 5 = 8.2 \text{ ms}$$

Note:- Priority Scheduling is Either Preemptive or Non-preemptive

Preemptive Priority Scheduling

Sr. No	Process	Burst time	Priority	Arrival Time
1	P ₁	10	3 (L)	0
2	P ₂	5	2	1
3	P ₃	2	1 (H)	2

Gantt Chart:



Priority Scheduling Algorithm

Completion Time:

$P1 = 17, P2 = 8, P3 = 4$

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT)

Turnaround Time(TAT) of each process

$P1 : 17 - 0 = 17 \text{ ms}$

$P2 : 8 - 1 = 7 \text{ ms}$

$P3 : 4 - 2 = 2 \text{ ms}$

Average Turnaround Time(ATAT) = Total turn around time of all processes / Total no of processes

Average Turnaround Time(ATAT) = $(17+7+2) / 3 = 8.6 \text{ ms}$

Priority Scheduling Algorithm

Waiting Time(WT) = Turnaround Time(TAT) – Burst Time(BT)

Waiting Time(WT) of each process

P1 : $17 - 10 = 7$ ms

P2 : $7 - 5 = 2$ ms

P3 : $2 - 2 = 0$ ms

Average Waiting Time(AWT) = Total waiting time of all processes / Total no of processes

Average Waiting Time(AWT) = $(7+2+0) / 3 = 3$ ms

Note:-

- if the two process having the same priority then process with shorter burst time will be executed.
- If the process having the same burst time then process will be executed on the basis of FCFS scheduling.

Starvation and Ageing

- **Starvation** or indefinite blocking is phenomenon associated with the Priority scheduling algorithms, in which a process ready to run for CPU can wait indefinitely because of low priority.
- To avoid starvation, we use the concept of **Aging**. In Aging, after some fixed amount of time quantum, we increase the priority of the low priority processes. By doing so, as time passes, the lower priority process becomes a higher priority process.

Round-Robin Scheduling Algorithm

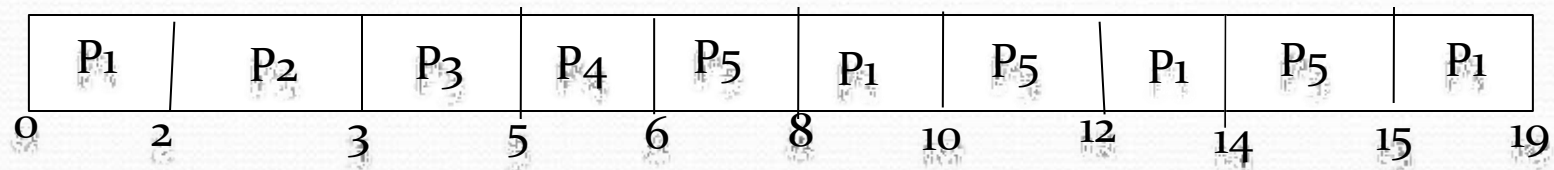
- It is designed especially for time sharing systems.
- Here CPU switches between the processes.
- When the time quantum expired, the CPU switched to another job.
- A small unit of time, called a time quantum or time slice.
- A time quantum is generally from 10 to 100 ms. The time quantum is generally depending on OS.
- Here ready queue is a circular queue.
- CPU scheduler picks the first process from ready queue, sets timer to interrupt after one time quantum and dispatches the process.

Round-Robin Scheduling Algorithm

Sr. No	Process	Burst time
1	P ₁	10
2	P ₂	1
3	P ₃	2
4	P ₄	1
5	P ₅	5

Time Quantum is 2 ms.

Gantt Chart:



Round-Robin Scheduling Algorithm

Completion Time:

P1 = 19, P2 = 3, P3 = 5, P4 = 6, P5 = 15

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT)

Turnaround Time(TAT) of each process

P1 : $19 - 0 = 19$ ms

P2 : $3 - 0 = 3$ ms

P3 : $5 - 0 = 5$ ms

P4 : $6 - 0 = 6$ ms

P5 : $15 - 0 = 15$ ms

Average Turnaround Time(ATAT) = Total turn around time of all processes / Total no of processes

Average Turnaround Time(ATAT) = $(19+3+5+6+15) / 5 = 9.6$ ms

Round-Robin Scheduling Algorithm

Waiting Time(WT) = Turnaround Time(TAT) – Burst Time(BT)

Waiting Time(WT) of each process

$$P1 : 19 - 10 = 9 \text{ ms}$$

$$P2 : 3 - 1 = 2 \text{ ms}$$

$$P3 : 5 - 2 = 3 \text{ ms}$$

$$P4 : 6 - 1 = 5 \text{ ms}$$

$$P5 : 15 - 5 = 10 \text{ ms}$$

Average Waiting Time(AWT) = Total waiting time of all processes / Total no of processes

$$\text{Average Waiting Time(AWT)} = (9+2+3+5+10) / 5 = 5.8 \text{ ms}$$

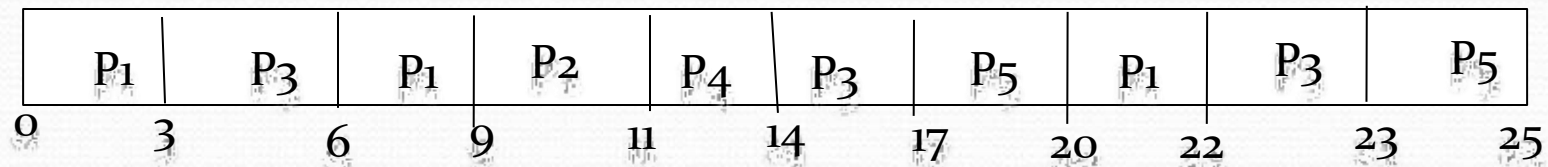
Round-Robin Scheduling Algorithm

when arrival time is given

S.No.	Process	Arrival Time (AT)	Burst Time (BT)
1	P ₁	0	8
2	P ₂	5	2
3	P ₃	1	7
4	P ₄	6	3
4	P ₅	8	5

Time Quantum (TQ)= 3 ms.

Gantt Chart:



Round-Robin Scheduling Algorithm

Completion Time:

P1 = 22, P2 = 11, P3 = 23, P4 = 14, P5 = 25

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT)

Turnaround Time(TAT) of each process

P1 : $22 - 0 = 22$ ms

P2 : $11 - 5 = 6$ ms

P3 : $23 - 1 = 22$ ms

P4 : $14 - 6 = 8$ ms

P5 : $25 - 8 = 17$ ms

Average Turnaround Time(ATAT) = Total turn around time of all processes /
Total no of processes

Average Turnaround Time(ATAT) = $(22+6+22+8+17) / 5 = 15$ ms

Round-Robin Scheduling Algorithm

Waiting Time(WT) = Turnaround Time(TAT) – Burst Time(BT)

Waiting Time(WT) of each process

$$P1 : 22 - 8 = 14 \text{ ms}$$

$$P2 : 6 - 2 = 4 \text{ ms}$$

$$P3 : 22 - 7 = 15 \text{ ms}$$

$$P4 : 8 - 3 = 5 \text{ ms}$$

$$P5 : 17 - 5 = 12 \text{ ms}$$

**Average Waiting Time(AWT) = Total waiting time of all processes /
Total no of processes**

$$\text{Average Waiting Time(AWT)} = (14 + 4 + 15 + 5 + 12) / 5 = 10 \text{ ms}$$

Advantages of Round-Robin Scheduling

- It doesn't face the issues of starvation effect.
- All the jobs get a fair allocation of CPU.
- It deals with all process without any priority
- This scheduling method does not depend upon burst time. That's why it is easily implementable on the system.
- Once a process is executed for a specific set of the period, the process is preempted, and another process executes for that given time period.
- It gives the best performance in terms of average response time.

Disadvantages of Round-Robin Scheduling

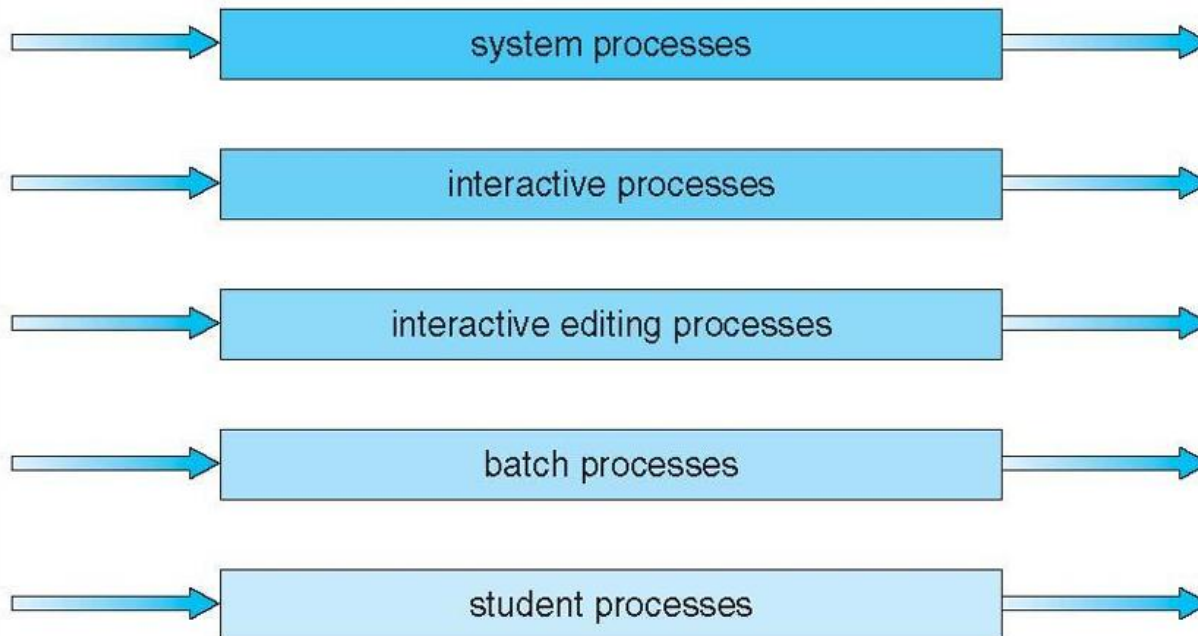
- If slicing time of OS is low, the processor output will be reduced.
- This method spends more time on context switching
- Its performance heavily depends on time quantum.
- Priorities cannot be set for the processes.
- Round-robin scheduling doesn't give special priority to more important tasks.
- Lower time quantum results in higher the context switching overhead in the system.

Multilevel Queue

- Ready queue is partitioned into separate queues, eg:
 - **foreground** (interactive)
 - **background** (batch)
- Process permanently in a given queue
- Each queue has its own scheduling algorithm:
 - foreground – RR
 - background – FCFS
- Scheduling must be done between the queues:
 - Fixed priority scheduling; (i.e., serve all from foreground then from background). Possibility of starvation.
 - Time slice – each queue gets a certain amount of CPU time which it can schedule amongst its processes; i.e., 80% to foreground in RR
 - 20% to background in FCFS

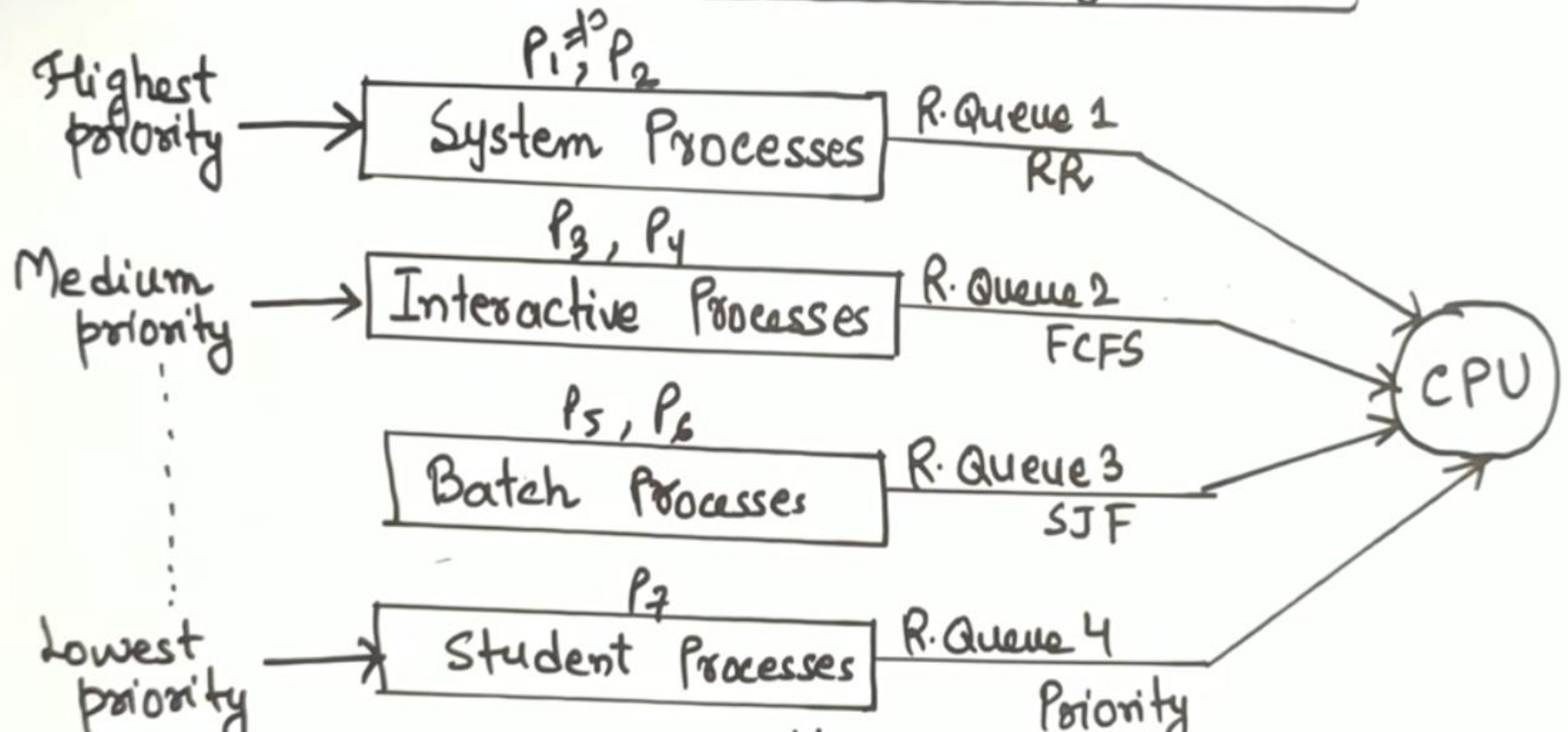
Multilevel Queue Scheduling

highest priority



lowest priority

Multilevel Queue Scheduling (MLQ)



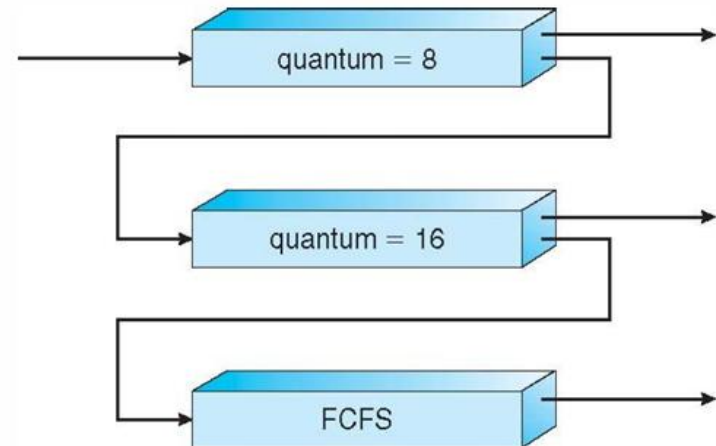
Example of Multilevel Feedback Queue

■ Three queues:

- Q_0 – RR with time quantum 8 milliseconds
- Q_1 – RR time quantum 16 milliseconds
- Q_2 – FCFS

■ Scheduling

- A new job enters queue Q_0 which is served ~~FCFS~~ **RR**
 - ▶ When it gains CPU, job receives 8 milliseconds
 - ▶ If it does not finish in 8 milliseconds, job is moved to queue Q_1
- At Q_1 job is again served FCFS and receives 16 additional milliseconds
 - ▶ If it still does not complete, it is preempted and moved to queue Q_2



Multi level feedback Queue Scheduling (MLFQ)

Highest
Priority

→ Queue 1 FCFS → Complete and exit

→ Queue 2 TQ = 4 → If complete then exit

→ Queue 3 TQ = 3 → If complete then exit

→ Queue 4 TQ = 2 → If complete then exit

Lowest
priority

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Thank You